

Hishaam Ali Abbasi

hishaamabbasi@gmail.com | 07710 812020 | hishaamabbasi.co.uk | [LinkedIn](#) | [GitHub](#)

PROFILE

Final year BEng Robotics Engineering student at Queen Mary University of London (IMechE accredited | Predicted First Class). End-to-end engineering experience across embedded firmware, mechanical design, control systems, SLAM, and DevOps infrastructure. Proven ability to take systems from concept to working prototype from a high-fidelity medical training device to an autonomous TurtleBot3 deployment. Strong practical builder with a full portfolio of documented case studies at hishaamabbasi.co.uk. Seeking a graduate role in robotics, embedded systems, or product development.

EDUCATION

BEng Robotics Engineering Predicted First Class

Queen Mary University of London | IMechE Accredited | 2022–2026

Key modules: Numerical Methods & Data Science (86%) | Control Systems & Analysis (78.9%) | Modelling & Control of Robotic Systems (75.2%) | Introduction to AI (74.7%) | Instrumentation & Measurements (76.1%)

Sutton Grammar School A Levels: Mathematics (A) | Physics (A) | Computer Science (B) | 12 GCSEs including Maths (9), CS (8), Physics (8)

EXPERIENCE

Engineering Design and Research Intern | Alvis Robotics, Birmingham — EOD Systems | August 2025 (2 weeks)

- Diagnosed and resolved a firmware fault on a proprietary sensor and camera activation module integrated onto a legacy Northrop Grumman EOD UGV platform. Traced to incompatible driver boards in the lighting and camera subsystem; restored full operational functionality by replacing the driver boards.
- Identified and resolved a mechanical failure on a tracked EOD platform where mud ingress had seized the active pitch mechanism on the crawler drive system. Performed full disassembly, fault diagnosis, and clearance to return the unit to operational condition.
- Supported R&D planning alongside the Managing Director on next-generation EOD robot development.

Additional experience: Clinical Data Administrator, AT Medics (2021); COVID Clinic Front of House, AT Medics (2020–2021).

PROJECTS

Fracture Immobilisation Trainer Final Year Dissertation ESP32 | Haptics | Fluidics | Sensors | 3D Printing | 2025–2026

- Led anatomical core workstream on a 5-person team; designed and manufactured a 1:1 adult-scale forearm chassis (NASA 50th-percentile anthropometric data) in Blender, incorporating internal routing channels for fluidics, sensor wiring, and haptic actuators.
- Ran a cantilever deflection study across four FDM materials under progressive loading. Selected PETG-CF on elastic stability grounds (apparent Young's modulus flat across load range, no plastic yielding vs 15% drift in PLA).
- Implemented dual-durometer silicone soft tissue system (Dragon Skin 30A muscle layer + Ecoflex 00-30 skin layer) informed by tissue mechanics literature; integrated Hall-effect magnetometers at the fracture site for real-time alignment tracking via ESP32.
- Delivered functional prototype for under £500 vs £2,000 commercial comparator (Limbs & Things Colles' Fracture Trainer), closing the feedback loop that passive commercial trainers lack.

TurtleBot3 SLAM and Autonomous Navigation ROS | SLAM | Gazebo | Python | Raspberry Pi | EMS627U Coursework

- Implemented differential drive kinematics and ROS-based SLAM using Gmapping, particle filter fusing LiDAR scan data with wheel odometry to build a real-time 2D occupancy grid map on a physical TurtleBot3 platform.
- Developed custom Python obstacle avoidance node replacing passive built-in behaviour LiDAR distance thresholding triggering autonomous reactive turns; successfully migrated full stack from Gazebo simulation to physical hardware.
- Resolved ROS/ROS2 package conflicts and OS dependency issues. Performed fresh Ubuntu 20.04 deployment on Raspberry Pi and configured a dedicated private network to bypass university Wi-Fi port restrictions on ROS peer-to-peer communication.

Formula Student Suspension System Redesign Ansys FEA | CAD | Vehicle Dynamics | Queen Mary Racing, Oct 2023–Present

- Fully redesigned 8 critical suspension components including wishbones, uprights, and bell crank for the new car platform using Ansys FEA.
- Optimised system geometry to achieve a 12% reduction in unsprung mass while maintaining peak structural stiffness, directly contributing to a targeted 0.3-second improvement in simulated lap time.
- Collaborated cross-department (powertrain, aerodynamics, chassis) to deliver a cohesive high-performance vehicle design.

Home Server and Self-Hosted Infrastructure Linux | Docker | Proxmox | Cloudflare | AWS | 2025–Present

- Built a production Proxmox VE hypervisor on a Lenovo ThinkStation (Xeon E3-1230v5, 16GB RAM, 9TB, RTX 3050) with Nvidia GPU passthrough to LXC container for NVENC hardware transcoding — resolved kernel pinning, cgroup device passthrough, and driver version conflicts.
- Deployed containerised services via Docker Compose: Jellyfin (4K/HDR), Nextcloud (8TB for 4 users), Nginx Proxy Manager, Portainer, and Duplicati with encrypted nightly backups to AWS S3 Glacier Deep Archive.

- Configured Cloudflare Tunnel and Zero Trust Access across 5 custom domains; eliminated open ports; migrated 4 websites from Vercel to self-hosted Nginx.

Active Camera Gimbal C++ | PID Control | Embedded

- Built a 2-axis gyroscopically stabilised camera gimbal with custom C++ firmware and real-time PID control loops on an embedded microcontroller.

Additional projects: 3D printed sim racing pedals (load cell braking, PETG-CF), Ender 3 V3 SE upgrade programme (Klipper, linear rails, remote monitoring), BoxTurtle multi-material filament system, robotic arm, web development, e-commerce shop. Full case studies at hishaamabbasi.co.uk.

LEADERSHIP

Explorer Scout Leader | Scouting Movement (UK) | 2012–Present

- Achieved Chief Scout Diamond Award, the highest level in Scouting. Founded a new Explorer section from inception; plan and deliver weekly sessions, expeditions, and residential camps for 18+ young people.

Flight Sergeant | Combined Cadet Force | Sep 2018–Jul 2023

- Progressed to highest NCO rank; taught the CCF syllabus, led teams on residential camps, and mentored younger cadets across varied environments.

TECHNICAL SKILLS

Programming: Python (Advanced) | Java (Advanced) | C++ (Intermediate) | C (Basic) | SQL (Basic) | HTML/CSS/JavaScript (Basic)

Robotics & Control: ROS / ROS2 | SLAM (Gmapping) | PID control | Kinematics | Gazebo simulation | Haptics | Sensor fusion | ESP32 | Arduino | Raspberry Pi

CAD & Manufacturing: Onshape (Advanced) | SolidWorks (Intermediate) | Creo (Intermediate) | Inventor / Fusion 360 (Intermediate) | Ansys FEA (Basic) | AutoCAD (Basic) | FDM 3D Printing (Advanced) | Orca Slicer (Advanced) | Klipper

Infrastructure: Linux (Debian / Ubuntu) | Proxmox VE | Docker / Docker Compose | LXC | Nginx | Cloudflare Tunnel / Zero Trust | AWS S3 / Glacier | systemd | Git

Other: Full UK driving licence | Native English | Conversational Urdu